# COMPUTER SCIENCE FORM ONE QUESTIONS

## Section 1: Questions based on Introduction to Computer Science

**1.** What are the key concepts and ideas that form the foundation of Computer Science? Describe these concepts.

**2.** Which field of Computer Science has caught your interest? Explain why.

**3.** Consider a potential profession in which you are interested. How do you believe Computer Science skills would be useful in achieving this?

**4.** In your own words, explain the importance of Computer Science.

**5.** Write a short essay on how Computer Science has reformed operations and efficiency in education and agriculture. Provide specific examples to support your answer.

**6.** Computer Science is only applicable to the technology sector. Discuss.

**7.** Describe the relationship between Computer Science and IT and how they complement each other in the modern technology landscape.

**8.** Explain other fields related to Computer Science apart from those mentioned in this chapter.

**9.** Elaborate on the collaboration between Computer Engineering and Computer Science in creating of computing systems.

**10.** Do you think it is important to learn the fields related to Computer Science? Briefly explain.

**11.** Explain the role of Computer Science in developing practical solutions to enhance transportation systems.

**12.** Explain the significance of Computer Science various in industries.

**13.** There has been a revolution in entertainment due to advances in Computer Science. With examples, explain how Computer Science plays a role in this revolution.

**14.** Discuss how Computer Science improves teaching and learning and give examples of software or platforms that support education delivery.

## Section 2: Questions based on Computer Systems

**1.** What if computers did not exist in today's world?

**2.** Baraka is a student at Kantalamba Secondary School. While in the school's computer laboratory, he enjoys playing games on his computer. Neema is the technician in charge of the computer lab. Describe their roles in terms of computer systems.

**3.** Is there any link between one computer generation and the next one? Explain.

**4.** What drove the transition from one generation of computers to the next?

**5.** Do you think computer generations have reached their end and we won't have another generation? Explain.

**6.** Compare and contrast the different classifications of computers and discuss the advantages and disadvantages of each type.

**7.** Give an instance of a task that necessitates the use of a supercomputer.

**8.** Give three reasons why a mobile phone is regarded as a computer.

**9.** Explain why we need different types of computers in today's world.

## Section 3: Questions based on Computer hardware

**1.** Briefly explain some of the considerations to be made when choosing an input device for a particular task.

**2.** Use various sources to explore the emerging field of 3D scanning and its applications in various industries, including architecture, engineering, manufacturing, and healthcare.

**3.** Differentiate between keyboards and mice as input devices.

**4.** What is the primary function of a CPU in a computer, and how does it execute instructions?

**5.** Explain the role of the motherboard in a computer system and how it connects various hardware components.

**6.** Provide examples of key components typically found on a motherboard and their functions.

**7.** Explain the differences between primary storage and secondary storage (auxiliary storage) in a computer.

**8.** Outline examples of common auxiliary storage devices and discuss their respective advantages and disadvantages.

**9.** Describe the purpose and operation of cache memory in a computer.

**10.** What is the difference between volatile and non-volatile memory in a computer system?

**11.** What are the advantages and disadvantages of using solid-state drives (SSDs) compared to hard disk drives (HDDs) for data storage?

**12.** What is the importance of registers in computer processing?

**13.** Differentiate between RAM and ROM and explain their importance in computer systems.

**14.** What is the primary function of an output device in a computer system? Provide examples of commonly used output devices.

**15.** What are the advantages of LCD screens over CRT displays?

**16.** Explain the difference between a plotter and a printer as output devices. What are the key characteristics and typical uses of each?

**17.** What is the role of speakers as output devices in a computer system?

## Section 4: Questions based on Computer software

**1.** Describe the importance of computer software in relation to Computer Science.

**2.** Differentiate between system software and application software.

**3.** Describe the steps for installing and uninstalling software.

**4.** What would happen if the computer we have today could not have (run) software?

**5.** Explain the relationship between hardware and software in a computer system.

**6.** How does the choice of software impact the functionality of a computer system?

**7.** Describe the importance of drivers and compatibility when connecting peripheral devices to a computer.

**8.** Explain why using drivers specifically designed for the operating system is crucial.

## Section 5: Questions based on computer system handling and care

**1.** Explain how to handle the effects of humidity on a computer.

**2.** Which computer parts might be affected if a computer is not properly maintained for a long time?

**3.** The computer is an electronic device that can be affected by power fluctuation. With an example, discuss this statement.

**4.** What is the significance of exercising hardware handling measures?

**5.** Explain how a voltage stabiliser protects a computer.

**6.** What is the difference between UPS and a voltage stabiliser in protecting a computer system?

**7.** Why is it advised to update your computer system regularly?

**8.** Computer malware is harmful to your computer system. Discuss.

**9.** Discuss possible risks a computer is likely to face if proper handling measures are not observed.

**10.** What potential risks or situations could arise if a computer user fails to back up their data regularly?

**11.** Describe how overheating can impact the computer system's performance.

**12.** Why is it important to keep operating systems and software applications up to date?

**13.** Describe the steps you would take to remove malware on a computer.

## Section 6: Questions based on Computer System maintenance

**1.** What are the potential consequences of neglecting computer system maintenance?

**2.** Why is wearing the Antistatic wrist strap said to be very important during computer maintenance?

**3.** What are the conditions that must be met when doing preventive maintenance?

**4.** Why it is important to dry RAM using isopropyl alcohol before inserting into the slot?

**5.** Why is it necessary to perform computer maintenance in a clean workspace?

**6.** What reasons, in your opinion, can necessitate replacing a RAM?

**7.** What circumstance can cause the Basic Input Output System to reset on your computer system?

**8.** Why is it necessary to reconnect the data and power cables after replacing a hard disk?

**9.** Assume you are working in a very busy office that produces a lot of data daily. Explain why and when you will use preventive, corrective, and routine maintenance.

**10.** Explain what can happen to a computer system if the disk is not regularly checked.

**11.** Why are disk cleanup and defragmentation essential for maintaining the efficiency of a computer hard drive?

## Section 7: Questions based on Computer System troubleshooting

**1.** Imagine a scenario where a computer is overheating frequently. Identify potential causes of overheating and describe the preventive measures that can be taken to address this issue.

**2.** What makes a computer system owner decide to defragment a disk drive?

**3.** Imagine you are a computer technician at school. A student informs you that their computer is running slower than usual, and they are unable to save their work. What steps would you take to restore the computer functionality?

**4.** Describe the steps to resolve a noisy fan in a computer system.

**5.** A computer monitor displays distorted or no image at all. What steps would you follow to identify and resolve the problem?

**6.** If a printer is not responding to print requests, explain the steps you would take to resolve the problem.

**7.** How would you resolve the issue if an external hard drive is not showing up on a computer?

**8.** What are the reasons for conducting performance troubleshooting?

**9.** Why is it necessary to conduct electrical troubleshooting in computer systems?

**10.** What are the common hardware problems that can occur in a computer system?

**11.** What are the common software problems that can occur in a computer system?

## Section 8: Questions based on problem-solving

**1.** What is the importance of studying problem-solving concepts in day-to-day activities?

**2.** Identify a sample problem and describe how to find a solution for that problem.

**3.** How can you apply problem-solving skills in other areas apart from Computer Science?

**4.** Define algorithm using the understanding you have gained from this section.

**5.** Why designing efficient algorithms very important in solving a problem?

**6.** Explore the school environment and find out any problem that may require an algorithm to solve.

**7.** Ndiyune has been given an unsorted list of numbers and his goal is to develop an algorithm to arrange these numbers in ascending order.

(a) Identify the problem that Ndiyune is supposed to tackle.

(b) Create an algorithm to solve a problem identified in (a).

**8.** Mwalimu Masiku is planning to distribute a set of "peremende" to a group of children. Each child must receive the same number of "peremende". Mwalimu Masiku intends to distribute the maximum number of peremende possible.

(a) Identify the problem that Mwalimu Masiku is supposed to solve.

(b) Write an algorithm to find the maximum number of peremende that Mwalimu Masiku can distribute to each child, ensuring an equal distribution.

**9.** Describe the meaning of flowchart as used in the representation of the algorithm.

**10.** Do you think, to this point, you can tell someone that flowcharts are not useful in problem-solving? Explain.

**11.** Is it possible that all symbols for the flowchart outlined in this section can be used in all types of problems that needs a solution?

**12.** Draw a flowchart to calculate the area of a circle.

**13.** Write an algorithm for withdrawing money from your bank account using an ATM card.

**14.** Provide an example of an algorithm for a task of your choice (e.g., cleaning your room, washing your clothes, cooking a porridge for your young sister, preparing for your midterm test) and explain the step-by-step process.

**15.** Design an algorithm for planning your school's Parents' Day, considering factors such as time, budget, and available resources.

**16.** Write an algorithm to calculate the area of a square.

**17.** What is pseudocode?

**18.** Can you tell someone that presenting a problem in pseudocode is useful? Explain.

**19.** Write pseudocode to calculate the sum and average of numbers.

**20.** Write a pseudocode for converting temperature from Fahrenheit into Celsius. The inputs to the algorithm are temperature in Fahrenheit, and the expected output is temperature in Celsius.

**21.** Write an algorithm to find the largest of three numbers: a, b, and c.

**22.** Why are algorithms crucial in problem-solving and Computer Science.

**23.** Why are algorithms required to be finite?

**24.** What is the importance of clearly understanding the problem before constructing an algorithm?

**25.** Write an algorithm that accepts four numbers as input and find the largest and smallest of them by using pseudocode and flowchart.

**26.** Write an algorithm to find a greater number between two different numbers entered by the user in an algorithm.

**27.** Write a pseudocode that reads two numbers, divides one by another, and display's the quotient. If the divisor is zero, display an appropriate message.

**28.** Your class teacher has adopted a program for computing final grades written by Form Two students. However, after using it for some time, the class teacher realised the program produces outputs that are not consistent. What type of problem is this? How can it be resolved?

**29.** Your best friend has shared their favourite computer game with you. Having realised that the game requires more computer resources than your computer has, you consider upgrading your computer so that you can install and enjoy the game. Having installed the game, you started playing it, but there was an error, and the game did not start successfully. The input devices did not work properly. What do you think should be done to play the game successfully?

**30.** John's office computer is running very slowly, he wants you to help him. What steps would you take to resolve the issue?

**31.** Suppose you want to use your computer, but it doesn't boot up properly when you switch on. How would you diagnose and resolve the booting problem?